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CHAPTER

Energy Resources

Chapter Outline

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INTRODUCTION

The term 'energy', the capacity to do work, has been coined by Thomas Young (1737-1829) and is applied to what it is now called kinetic energy. The two laws of thermodynamics describe the behaviour of energy. The first law states that energy can be transformed from one form to the other, but cannot be created or destroyed. The second law states that some energy is always dispersed into unavailable heat energy. No spontaneous transformation of energy from one form to another (in context of protoplasm) is 100 per cent efficient.

Energy is an essential input for the economic development and improvement of quality of life. Every work in this world is accomplished with the help of energy. Infact, all living beings operate by means of energy. Energy moves the universe. About 99.8 per cent of our energy comes from solar radiation. It is this solar energy that plants use to make food which gets stored in plants as biomass. Other energy inputs to the earth are wind energy, ocean thermal and tidal energy, geothermal energy, nuclear energy and energy from fossil fuels, such as oil, coal and natural gas. Many other alternate sources of energy are being explored and used.

About 24 per cent of energy consumed globally, is used for transportation, 40 per cent for industries, 30 per cent for domestic and commercial purposes

and the rest 6 per cent for other uses including agriculture. About 2 billion people, one third of global population living in developintg countries, lack access to adequate energy supplies. Three billion people depend on fuel wood, coal, charcoal, dung and kerosene *etc.* for cooking and heating. On the other hand, industrialised nations, with only 25 per cent of global population, account for 70 per cent of commercial energy consumption. Most of it comes from oil that is used for transportation. The United States, with just 4.6 per cent of world's population, consumes 24 per cent of the total commercial energy produced and is the largest consumer of energy in the world. India, with 16 per cent of world's population, accounts for just 3 per cent of the total energy consumption of which about one-third comes from biomass. There is a wide disparity in energy consumption in developed and industrialised countries and developing countries. The energy consumption of a nation now-a-days is usually considered as an index of its development.

Today, energy is the primary input for almost all economic activities and has become vital for improvement in the quality of life. Infact, the whole infrastructure rests upon energy. Use of energy in various sectors, such as industry, commerce, transport, telecommunication, agriculture and household services has compelled us to focus our attention to ensure its continuous supply to meet our ever increasing demands. The modern civilisation is much dependant on energy availability but the energy resources, mainly oil, coal, natural gas and hydroelectric power, are becoming scarce and costlier. The prices of the oil, the most common source of energy, are going very high. To cope with ever increasing demand, efforts are being made to develop new approaches in conventional and non-conventional energy sources and new measures of energy conservation.

In India, power development commenced at the end of 19th century with the commissioning of electricity supply in Darjeeling in 1897 followed by commissioning of a hydropower station at Sivasamundram in Karnataka during 1902. In pre-independence period the supply was in private sector that too was restricted to urban areas. After independence, with the formation of State Electricity Boards during Five Years Plans, a significant step was taken to bring about systematic growth of power supply in industries all over the country. National Electricity Policy, 2005 aims at accelerated development of power sector, providing electricity supply to all areas and protecting interests of consumers and other stakeholders with the view of availability of energy resources, technology available to exploit these resources, economics of generation using different resources and energy security issues.

RENEWABLE AND NON-RENEWABLE ENERGY SOURCES

Energy sources can be spread over renewable and non-renewable. The renewable energy sources are regenerated in natural process so that they can be used indefinitely. The renewable energy sources generally cause little or no environmental impact as compared to non-renewable sources, such as fossil fuels and nuclear energy. Keeping in view, the current state of technology, the generation of renewable energy is often more expensive than energy produced by fossil fuels or nuclear energy. However, with the technology advances, the cost of renewable energy is expected to decrease. Renewable energy sources are inexhaustible, *i.e.,* they can be replaced after we use them and can produce energy again and again. Among the renewable energy sources, the most commonly and easily accessible is solar energy. The other renewable energy sources are hydropower, wind, geothermal, ocean wave and tidal energy. Non-renewable energy sources include various fossil fuels including petroleum products, coal and natural gas and nuclear energy. Nuclear energy is mainly obtained from the nuclear fission of uranium or thorium. The global resources of fossil fuel and uranium and thorium are limited and will eventually be depleted. Moreover, use of fossil fuels for energy has negative environmental consequences, such as air pollution, global warming, acid rains and oil spills. Thus, it has become essential to minimise the use of fossil fuels and to replace them with renewable resources. Keeping in view the growing energy needs, use of non-coventional energy sources, over conventional, should be promoted.

RENEWABLE (NON-CONVENTIONAL) ENERGY SOURCES

Renewable energy sources are regenerated in natural processes and can be used indefinitely. These are available in great amount in nature and develop in a relatively short period of time. These include solar energy; wind energy; ocean thermal power; ocean wave and tidal energy; geothermal energy; hydropower *etc*. They are capable of solving the twin problems of energy supply in decentralised manner and helping in sustaining cleaner environment.

Solar Energy

Among renewable energy sources, the easily accessible and most important is the solar energy. It can be used directly or indirectly for human welfare. The direct solar energy is the radiant energy, whereas the indirect solar energy is the energy obtained from the materials, such as biomass, in which sun's radiant energy has been incorporated by the plants. Prof. H.P. Garg of I.I.T. Delhi is considered as Father of Indian Solar energy.

***(i)* Direct Solar Energy:** On global-scale, 15 days of solar energy is roughly equivalent to the energy stored in all known reserves of fossil fuels on the earth. The continuous input of the energy from the sun is 1,67,000 times greater than the current consumption. As such, now many countries are in effort to harness the solar energy for domestic, commercial and industrial purposes. Solar energy can be used for direct heating. Alternatively, the heat can be converted into the electricity—the thermal electric generation. Photovoltaic cells—the solar cells or solar batteries convert direct solar energy into electricity. Other types of solar energy devices are solar power pumps (Fig. 7.2) and solar pond electric power plants (Fig. 7.3). Direct utilisation of solar energy is also being tried by using part of natural oceanic area as a gigantic solar collector and constructing large-scale ocean thermal plants.

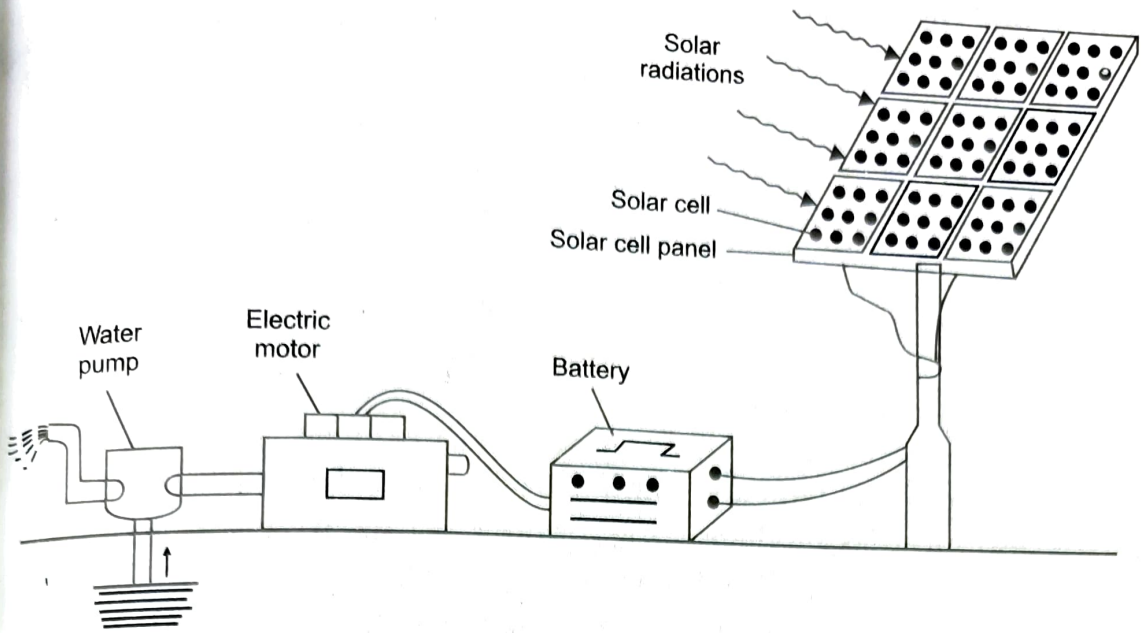


Fig. 7.2. A solar pump run by electricity produced by solar cells.

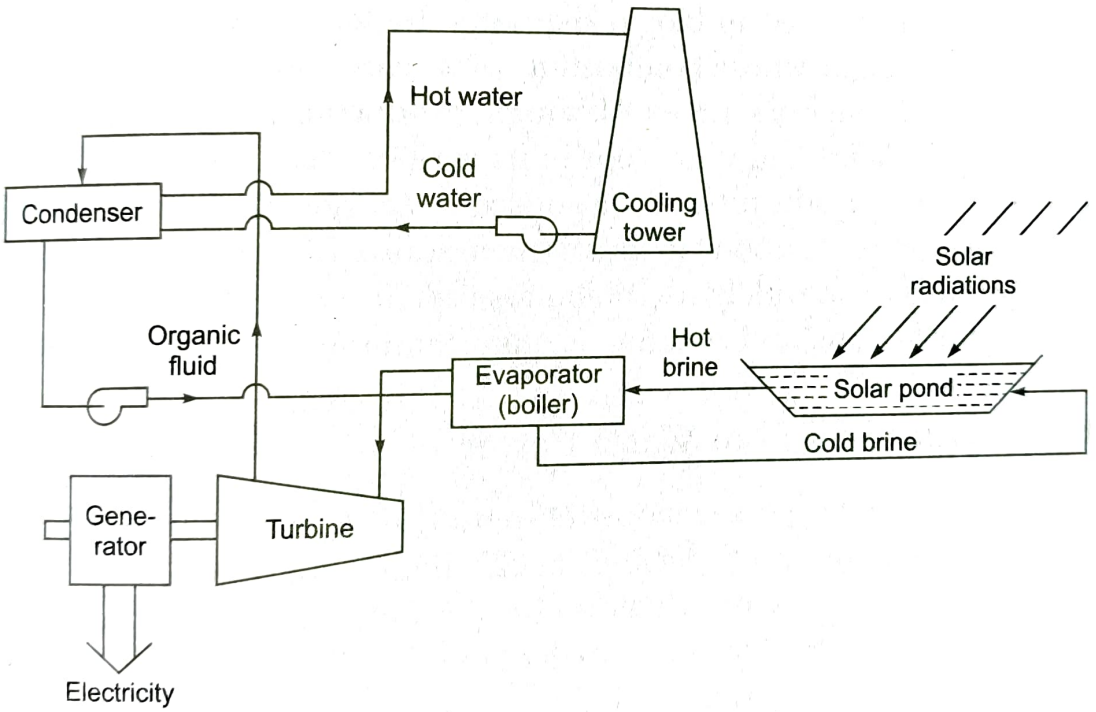


Fig. 7.3. Solar pond electric power plant.

The only need is to develop a massive-scale back-up system to store and generate electricity when solar power is not operative at night or during the cloudy days. Government of India has planned to set up **Akshay Urja shops** to cover every district of the country. These shops create outreach of renewable systems/devices. So far 269 Akshay Urja shops have been opened by 2009.

In tropical countries like India, the use of solar energy could prove most suitable in the long run as sunlight is available in plenty, almost throughout

the year. India has been one of the major producers of photovoltaic systems, currently used in Street Lighting Systems (SLS), Domestic Lighting Systems (DLS), Community Lighting Systems (CLS), water pumping systems, small power plants, solar cookers *etc.* India installed largest solar steam cooking system for cooking food for 15,000 people per day at Tirupati Temple in Andhra Pradesh.

***(ii)* Indirect Solar Energy (Biomass):** Biomass energy, where solar energy is utilised indirectly, has been the major source of energy to human beings throughout the history of civilisation. Biomass energy is the outcome of the photosynthesis and is bestowed in the materials, such as live plant material and their dried residues; fresh water and marine algae; and agricultural and forest residues of plant and animal origin. Apart from these materials, biomass also includes biodegradable organic wastes from industries like sugar mills, breweries *etc.* Biomass fuel may be solid, such as wood, animal dung, peat, charcoal *etc.* used for burning; liquid, such as methanol or ethanol used in internal combustion engines of automobiles; or gas, such as animal waste produced biogas: a mixture of gases mainly methane and some carbon dioxide produced in biogas digesters. Biogas is a clean anaerobic fuel of organic origin whose combustion produces fewer pollutants than other combustible energy sources. Although, production of energy requires sufficient area of land and water, but in its various forms, biomass energy appears to have a bright future as a source of energy. At least half of the global population relies upon biomass as main source of energy for domestic use. Firewood is most widely used biomass fuel in the world. In Scotland, use of peat for heating and cooking, is quite common.

Hydroelectric Energy or Water Power

Water power has been in use since the period of Roman Empire. Water falling from a height turns turbines at the bottom of dams to generate electricity (Fig. 7.4). Presently, water power is widely used all over the world and produces approximately one-fourth of the world's electricity. Electricity produced by hydro-power is much cheaper than electricity produced by thermal power. Water power is a clean energy source as it does not cause environmental pollution and the electricity produced can be transmitted to long distances through wires or cables (Fig. 7.5). But building a dam leads to several environmental problems, such as submergence of plant and animal habitats and displacement of local inhabitants including tribals.

About 75 per cent of the total electricity consumed in South America is derived from water-power. Japan, US and Russia are the leading countries in the production and consumption of hydroelectric power. Canada, Norway, New Zealand, Switzerland and many other European countries have trapped most of their water sources and harnessed energy to generate electricity.

Geothermal Energy

The natural heat from the interior of the earth can usefully be converted into energy. This natural heat comes from the fission of radioactive material present in the rocks in the interior of the earth. The idea of harnessing geothermal energy, utilising dry steam, was developed in Italy as early as in 1904. Natural internal heat of the earth was harnessed by geo-hydro-thermal conversion, hot igneous and geo-pressured systems. Presently, there are several geothermal plants working successfully in USA, New Zealand, Russia, Japan, Mexico and California. Heated groundwater flowing upwards as hot water stream and hot springs—the nature's geysers, can be used to turn turbines and generate electricity.

In India, natural geysers are common in Kullu and Manali, Manikarn, Sohana and some other places. Assessment of geothermal energy potential of selected sites in Jammu and Kashmir, Himanchal Pradesh, Uttarakhand, Jharkhand and Chhattisgarh is being undertaken. Some more field investigations, including deep drilling at potential geothermal sites, would be required before these sites can be taken up for development for geopower generation.